Data Science Fundamentals

Explaining Potential Errors in Simplified Representations of Data

An example of the Simpson's Paradox (**where a trend or pattern in the data disappears or reverses when combined with other data**) is if the bank *Bank of America* collected data from an extensive survey to determine which of its three main corporate loans were most popular.

Bank of America issues 2-year, 5-year, and 10-year corporate loans. The sample size collected was 100 companies.

Companies favored the 10-year loan (**80% favorability**), 5-year loan (**60% favorability**), and 2-year loan (**50% favorability**)

However, the paradox confronted was that other companies actually held more loans in total reverse order in sum to the favorability rate. **75 companies** had a 2-year loan, **15** had a 5-year loan, and **10 companies** had a 10-year loan.

The poor business decision was to sell more 10-year loans. This was based on looking at the **aggregated data** of the 10-year loan having a 80% favorability. The lurking variable was the number of loans held by companies (total reverse to the favorability rate). The aggregated data was misleading when taken at face value, and it is **always important to investigate further into the data to understand the whole picture**. In the context of Bank of America, the business decisions that need to be made need to take into account the larger picture, as more companies possessed loans that were not evaluated as being favorable in the bank's initial survey.

Assuming a managerial role, I would **reverse the business decision to sell more 10-year loans**, and instead create a *price-to-loan-ownership-rate business model* that would sell loans at **appropriate prices** based on the particular number of 10-, 5-, and 2-year loans held by companies. 2-year loans would become the most expensive, 5-year loans would become middle-range in cost, and 10-year loans would become the cheapest. This would be done to influence overall demand by companies to buy more loans that are less popular in terms of total numbers of ownership, in the hope of selling loans that are not selling as well.